

Unit 12, Lesson 7: Percent Increase and Decrease



Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.



Learning Target

☐ I can find percent increase or percent decrease.



12.7.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?

2. Explain your answer using mathematics.



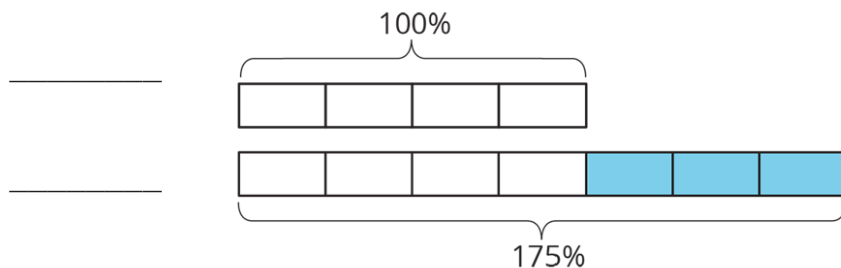
12.7.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

- It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

The diagram models this in terms of percentages.

- Label each line of the tape diagram with the name of the student who is represented.



- Complete the statements.

The time it takes Andre to get to school is 75%

- ☐ more
- ☐ less

than the time it takes Jada. The

diagram shows Andre's travel time is ☐ 75% ☐ 175%

of Jada's travel time.

- Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.

- If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.

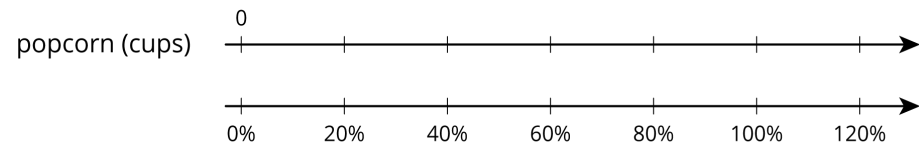
The terms "percent increase" and "percent decrease" are used to describe a specific **percent of change**.



A **percent of change** is the difference between a final value and an initial value, expressed as a percentage of the initial value.

A double number line can also be used to model percent increase or percent decrease.

- A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



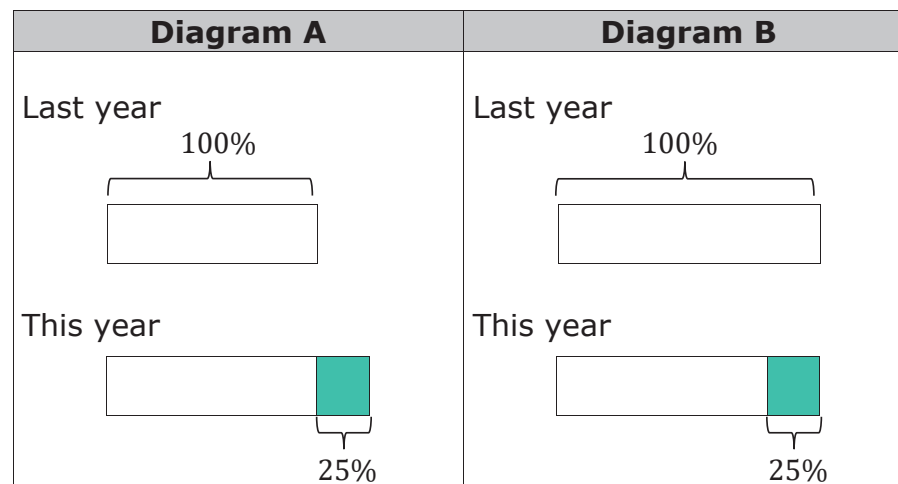
- On the top number line, plot the 15 cups of popcorn as the initial value of 100%.
- On the second line, plot 20% less than the initial 100%.

- c. Use the double number line to find the number of cups in the smaller bag.
- d. Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.



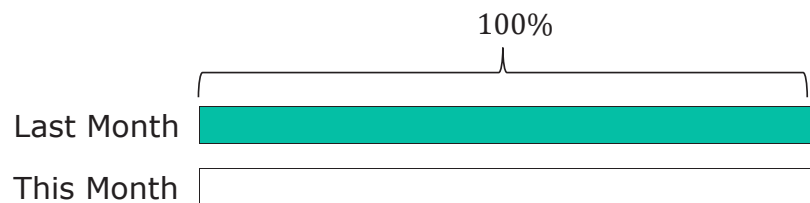
12.7.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

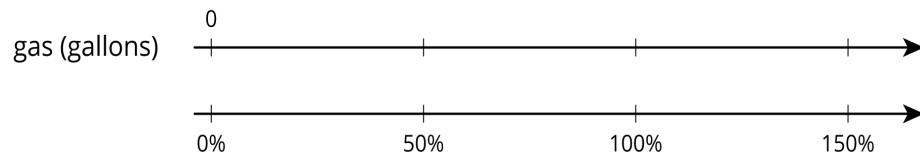


Situation	Matching Diagram	Reason
Compared to last year's strawberry harvest, this year's increased 25%.		
This year's blueberry harvest is 75% of last year's.		
Compared to last year, this year's peach harvest decreased by 25%.		
This year's plum harvest is 125% of last year's plum harvest.		

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



12.7.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains "20% More" cereal. The original cereal boxes contained 500 grams of cereal.
 - a. Explain how to determine how much cereal is in the new box.



- b. How much cereal comes in the new box?

2. The price of a shirt at ClothesMart is \$18.50, but there is a store coupon that lowers the price by 20%.
- a. Work with your partner to determine how much the shirt is with the coupon.



The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	Approach 2
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what's remaining after the 20% has been deducted from the original cost.
New cost = $0.8(18.50)$	New cost = $18.50 - 0.2(18.50)$

- b. Explain whether or not these equations are equivalent.

- c. Did your answer to Part A match the solution(s) of the equations in the table?



$$\text{Percent of Change} = 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

3. The original cost of pants at ClothesMart is \$30.00, and after a store coupon is applied, the cost is only \$25.50.

- a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{\boxed{} - \boxed{}}{\text{Initial}} = \frac{\boxed{} - \boxed{}}{\boxed{}} = \frac{\boxed{}}{30} = \boxed{}\%$$

- b. Explain whether the percent of change represents a percent increase or decrease based on its value.

- c. Complete the statements.

When calculating the percent of change using the ratio,

a positive value represents a percent ☐ increase,
☐ decrease,

and a negative value represents a percent

☐ increase.
☐ decrease. In this context, it makes sense that the

percent change is a percent ☐ increase
☐ decrease because the

change is the result of a discount.



12.7.5 Your Turn!

1. A school had 1,200 students last year and only 1,080 students this year. What was the percentage decrease in the number of students?
2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?



12.7.6 Wrap-Up: Error Analysis

1. The number of campers who attend a local summer camp has increased from 1,938 to 2,128. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

$$2,128 - 1,938 = 190$$

$$190 \div 2,128 = 0.089$$

$$0.089 = 8.9\%$$

Explain the camp director's mistake.

Unit 12, Lesson 8: Calculating Percent Error and Simple Interest



Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.



Learning Targets

- ☐ I can find percent error.
- ☐ I can solve problems that involve simple interest.